

MOHAMED ADEL MAHMOUD

Cairo, Egypt | +20 101 221 8724 | mohamedadelamin12@gmail.com | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Fresh Computer Science graduate focused on AI and cloud engineering, with hands-on experience building machine learning, computer vision, RAG, backend API, and cloud-deployed solutions.

EXPERIENCE & TRAINING

Backend Intern | Connect Digital Solutions

Aug 2025 - Nov 2025

- Developed Java Spring REST APIs and worked with professional API-oriented development workflows.

Data Science Program | ALX Africa

Mar 2025 - Apr 2026

- Completed analytical workflows, portfolio projects, and structured problem-solving practice.

Data Science Track | DEPI

Oct 2024 - May 2025

- Applied data analysis, machine learning, visualization, and practical project delivery.

EDUCATION

Arab Academy for Science, Technology & Maritime Transport

Bachelor of Science in Computer Science | Graduated: 2026 | GPA: 3.54

SELECTED PROJECTS

PharmaSafe: AI-Powered Medication Safety Platform | Graduation Project (A+)

- Developed a full-stack medication-safety platform using FastAPI, React, and Flutter, combining DDI detection, prescription OCR, patient-aware substitution, and safety reporting.
- Built a hybrid DDI engine using DrugBank/Neo4j and 90,543 DDInter records, with a PubMedBERT fallback achieving 88.14% accuracy and 85.63% macro-F1.
- Designed a drug-substitution pipeline that filters and ranks safer alternatives against the patient's medication profile and DDI evidence for pharmacist review.
- Implemented a static digital twin that models the patient's current medication state, compares original and substituted regimens, and produces explainable safety reports.

Intelligent Malware Analysis Platform

- Combined static ML, dynamic sandbox behavior, and VirusTotal hash checks; achieved XGBoost AUC 0.99 and accuracy 0.97 on EMBER2024.
- Deployed using Huawei Cloud ECS, OBS, VPC, and Cloud Eye.

RAG Document Q&A Assistant

- Built source-grounded Q&A for PDF, DOCX, and TXT files using LangChain, FAISS, embeddings, Groq LLMs, Gradio, and FastAPI.

Brain Tumor Detection System

- Achieved 97% MRI tumor-classification accuracy on 7,000 images using PyTorch and ResNet-18.

Disaster Damage Assessment from Satellite Images

- Classified post-disaster building damage from pre/post imagery with 78% accuracy and 76% macro-F1 using Swin Transformer-Tiny and SAM2.

CERTIFICATIONS & ACHIEVEMENTS

- AI / Generative AI

[IBM AI Engineering Professional Certificate](#)
[IBM Generative AI Engineering with LLMs Specialization](#)

- NVIDIA / ITI

[Generative AI - Beginner Level \(35 hrs.\)](#)
[Building LLM Applications With Prompt Engineering](#)
[Generative AI - Advanced Level \(100 hrs.\)](#)
[Building RAG Agents with LLMs](#)

- Oracle

[Oracle Cloud Infrastructure Certified Foundations Associate](#)
[Oracle Cloud Infrastructure Certified Architect Associate](#)
[Agentic AI Certified Foundations Associate](#)

- Huawei Cloud

[HCCDA-Tech Essentials](#)
[HCCDA-AI](#)

- Graduation Project Grade A+; Top 50 in Huawei Developer Competition Northern Africa 2025 among 389 teams and 1,400+ developers;
- ECPC 2024 Finalist; four academic course project awards.

SKILLS

- AI / ML: Python, PyTorch, TensorFlow, Keras, scikit-learn, CNNs, transfer learning, computer vision, transformers, model evaluation.
- Generative AI / RAG: LLMs, LangChain, FAISS, embeddings, prompt engineering, Groq LLMs, source-grounded QA.
- Cloud: Huawei Cloud, ECS, OBS, VPC, Cloud Eye.
- Frontend / Backend: Java, Spring Boot, FastAPI, REST APIs, React, Flutter, HTML, CSS.
- Tools: Neo4j, Gradio, Streamlit, Git, GitHub.